

Employee Attrition Prediction – Executive Summary

Prepared for: HR Director

This project analyzed employee information to identify the key factors that influence employee attrition and to help HR teams take preventive actions before valuable employees leave the organization.

The analysis found that employee turnover is not driven by salary alone. Employees working as Laboratory Technicians and Sales Representatives, as well as those who frequently travel for work or work overtime, are more likely to leave the company. Employees with lower work-life balance and shorter tenure at the organization also show a higher risk of attrition.

Among all departments, the Sales Department recorded the highest employee attrition rate, indicating that HR should prioritize retention efforts in this area. Although lower monthly income is associated with higher attrition, workplace conditions, job responsibilities, and career growth opportunities appear to have a greater impact on employees' decisions to leave.

Based on these findings, HR should introduce regular retention discussions for employees in high-risk roles, reduce unnecessary overtime wherever possible, improve work-life balance, and provide clearer career development and promotion opportunities. These initiatives can improve employee satisfaction and reduce voluntary resignations.

The predictive model developed during this project can help HR identify employees who may require additional support or engagement. However, it should be used as a decision-support tool rather than a replacement for human judgment, since personal circumstances and external job market conditions are not included in the available data.

Overall, this project demonstrates how employee data can be used to support proactive HR planning, improve retention strategies, and reduce the long-term costs associated with employee turnover.